Superresolution CT

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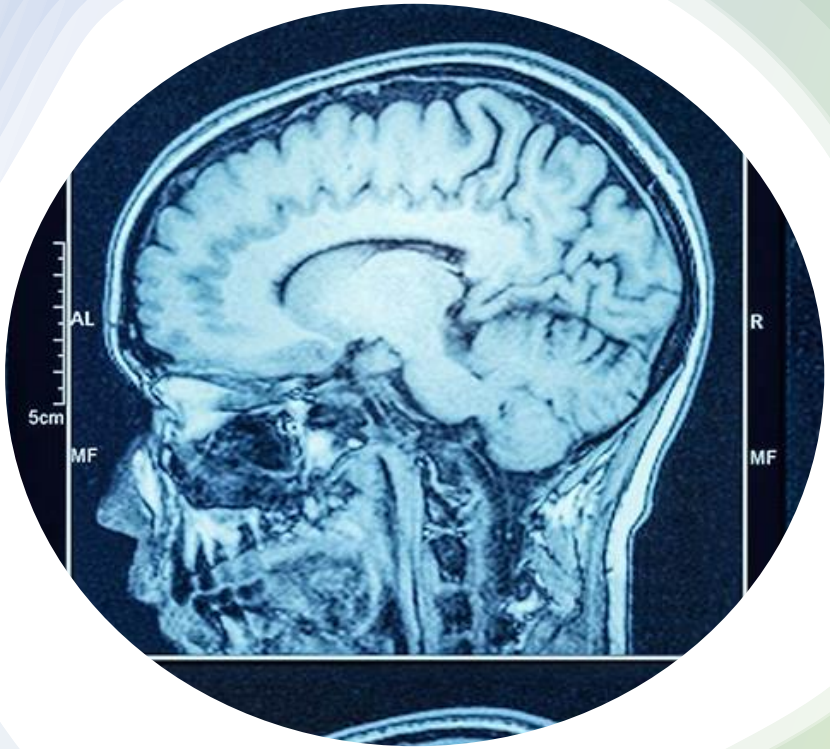


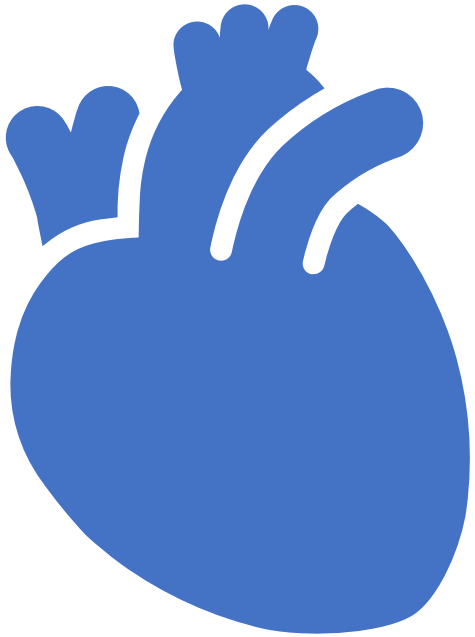
General Idea

- Obtaining very high-resolution from low resolution images (2.6 um voxel, compared to 0.2 mm)
- Small radiation dose
- UNet neural network (machine learning algorithm)

Medical Problem

- Detailed structures of organs
- **Research**: Myocardial Fibrosis

(CDC, 2023) (Yetman, 2023) (Heidenreich et al., 2022) (Bozkurt, 2023)



Market

- Cardiovascular disease research: **$0.44 trillion**
- E.g. Myocardial Fibrosis Market: **$1.6 billion**
- Heart failure cost: **$60 billion**
- Annual CT Scans: **70 million** in USA
- **3 million** pacemaker USA users
- Competition: **High Resolution MRI**

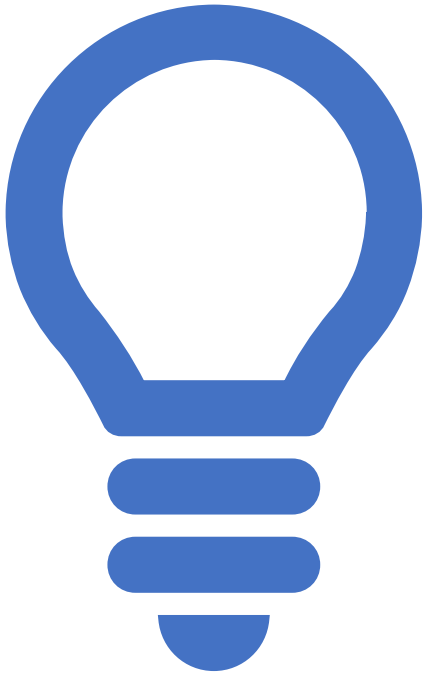
Potential Profit

- **Buyers**: Hospitals, Diagnostic Centres, Private Clinics
- 1 scan ~ **$300** per scan

(Bosch de Basea et al., 2015) (Imaging, 2017) (Schoenfield, n.d.) (Precedence Research, 2023) (Statista, 2023)

Uniquity

- Much higher resolution than normal CTs
- Research purposes
- Not limited to cardiovascular
- Easy to use
- Does not wear out
- Low dose



Achievements to date

- Readiness Level 2 – Early research stage
- Concept devised
- Research groups in place and funded

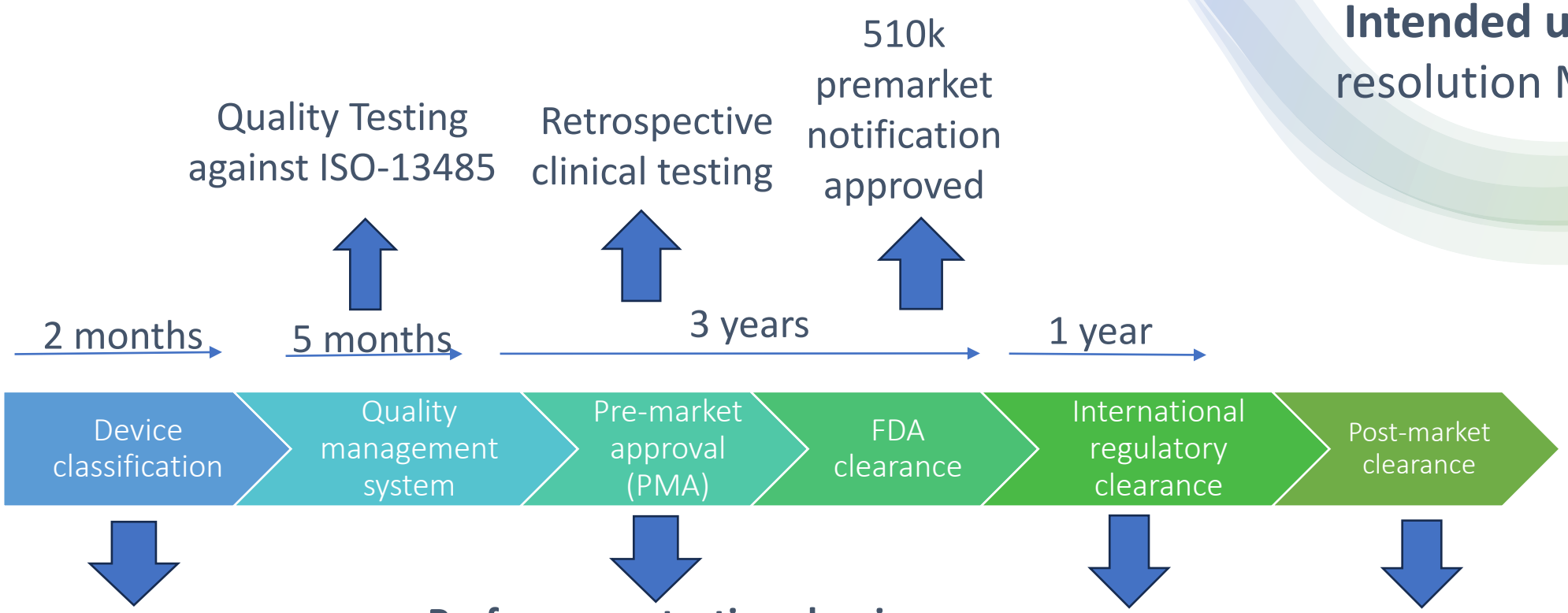
Intellectual Property

- Exists with **Adobe Inc** and **Taiyuan University** of Science and Technology
- Most of code is done on Public Domain
- **Solution 1**: Patent original parts of code
- **Solution 2**: Rely on copyright and secrecy
- **Conclusion**: Intellectual Property is uncertain

(Zhang, Lin and Wang, 2018) (Xiong Zhang et al., 2019) (Urzedowski, 2022)

Regulatory path

Comparator device: UMR 680
Intended use: High-resolution MRI scans



Class 2 Medical Device

Performance testing clearing requirements:

- Resolution: 2.6um (x1000)
- Image contrast resolution
- Noise and uniformity
- Image artifacts

Regulatory laws differ within EU and language translation

Monitor safety of device

(Antun et al 2020) (Fenton, 2021) (Science Business, 2023)

The Team

Year 1 - 4

- CEO
- Cybersecurity
- Software Engineers x 3

Year 5 - 6

- CEO
- Cybersecurity
- Software Engineers x 3
- Radiologists x 2

Year 7 - 9

- CEO
- Cybersecurity
- Software Engineers x 3
- Radiologists x 2
- Post Market Surveillance
- Legal Advisor
- Accounting Advisor
- Quality Regulatory Manager
- Sales x 2

(Salary.com, 2024)

Finance

Job	Salary /per year
CEO	$150,000
Cybersecurity	$80,000
Software engineer	$87,500
Radiologist	$33,000
Post market surveillance	$75,000
Legal advisor	$110,000
Accounting advisor	$63,000
Sales	$50,000
Quality regulatory manager	$103,000

Yearly gross salaries of employees:

- Year 1-3 salaries: $1,477.5k
- Year 4 salaries: $492.5k
- Year 5-6 salaries: $1,117k
- Year 7-9 salaries: $2,019k

Resources needed:

- Space: $50,000 (annual)
- Expenses: $20,000 (annual)

Total investment needed (years 4-9) :

- $4,048,500 ≈ **$4.1 million**

(Salary.com, 2024)

The Future

1-3 years

- Adapt technology to use in-vivo
- Improve the neural network (alive patients)

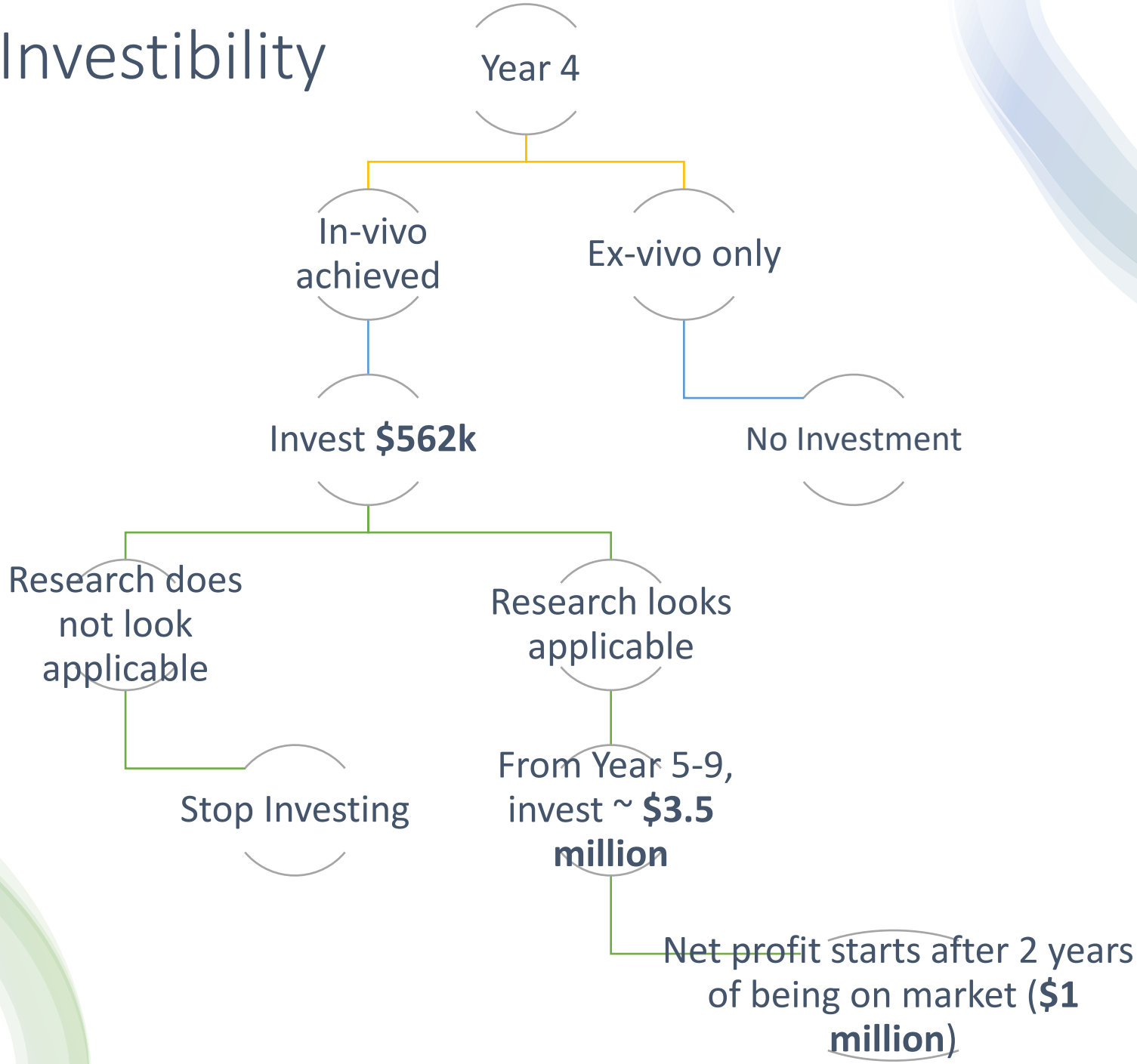
4-6 years

- Clinical Trials
- Clinical setting or Research only

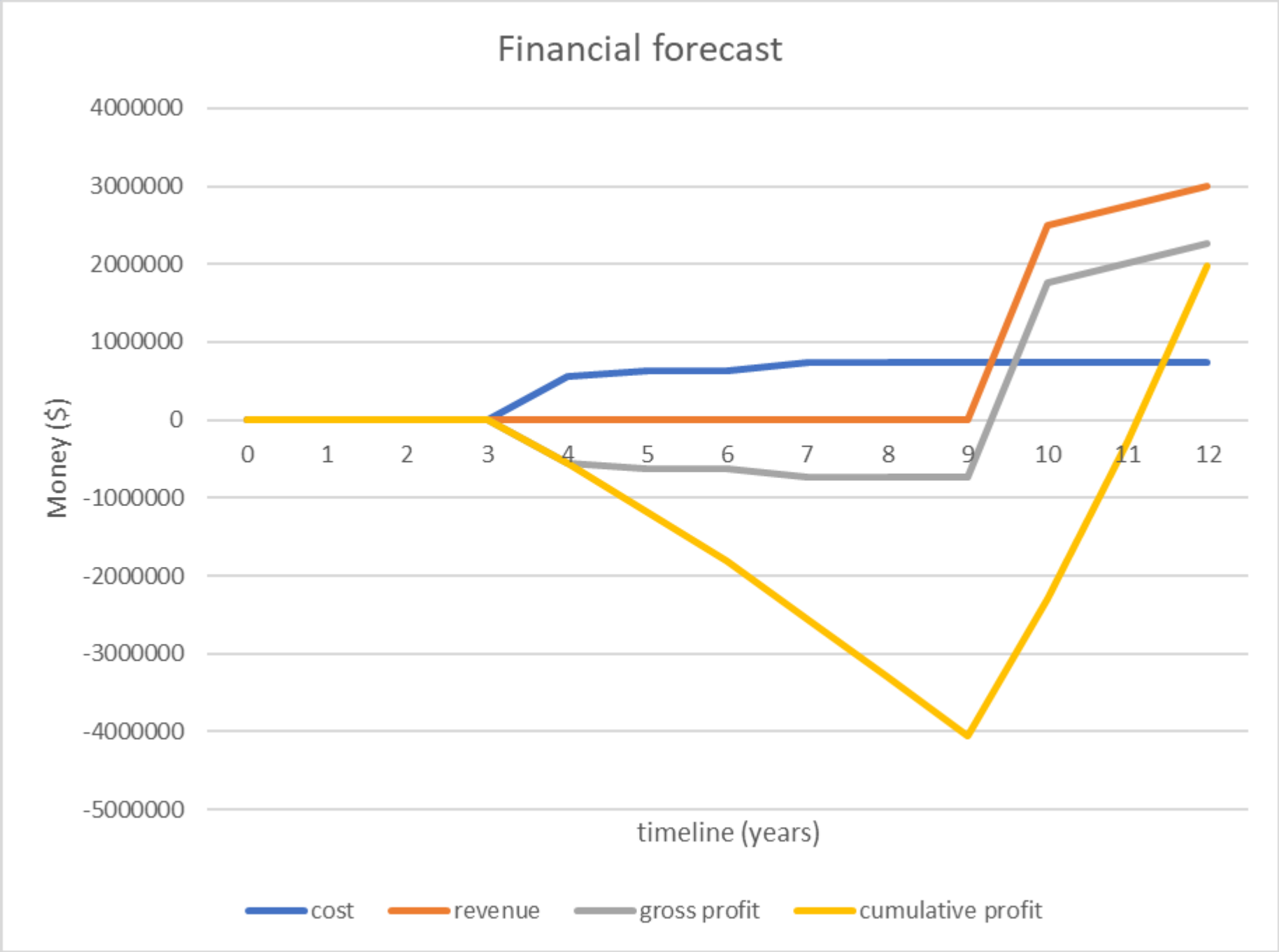
7-9 years

- Approval process
- Marketing

Analysis of Investibility



(Salary.com, 2024)


Financial forecast
4000000
3000000
2000000
1000000
0
-1000000
-2000000
-3000000
-4000000
-5000000
Money ($)
0
1
2
3
4
5
6
7
8
9
10
11
12
timeline (years)
cost
revenue
gross profit
cumulative profit

Conclusion

- Put a small amount of money in year 4 (partial investment)
- Based on the advancements made on research, invest more / stop investment